Customer Churn Prediction & Retention Analysis

Business report | End-to-end machine-learning solution

Dataset size: 1,200 records and 20 source columns. Churn prevalence: 86.3%.

Executive summary

This project builds a reproducible pipeline to rank customers by churn risk, compare three classifiers, and simulate a constrained retention campaign. The selected model by cross-validated average precision is **Random Forest**. On the untouched test set, PR-AUC is 0.981, ROC-AUC is 0.905, recall is 0.995, and precision is 0.862 at the training-selected threshold of 0.36.

Business interpretation

The assignment states that only a limited share of customers can be contacted. Therefore, ranking quality and recall/precision trade-offs matter more than accuracy alone. The simulation evaluates the top 20% of the held-out test set and compares it with random contact. Financial outputs are illustrative because the dataset does not provide campaign costs, retention conversion rates, or customer lifetime value.

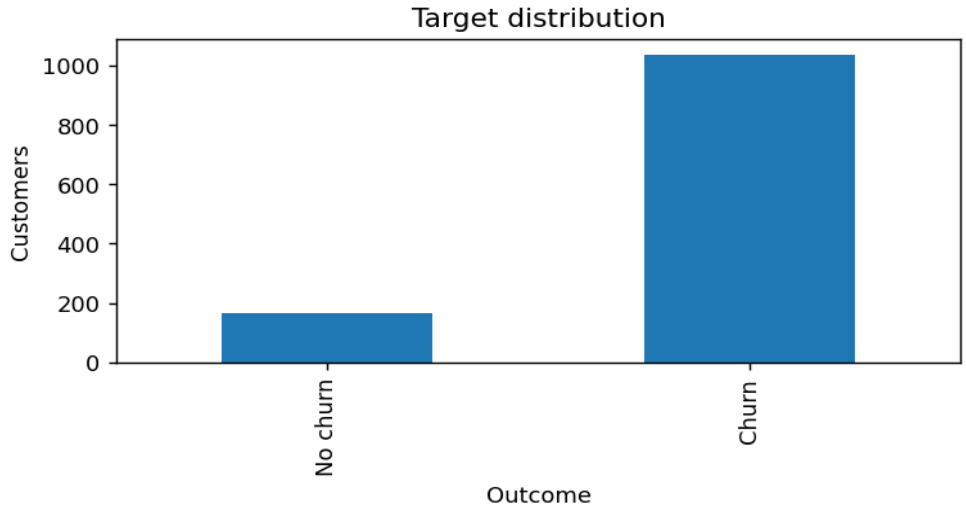
Important data caveat

The supplied data has a high churn prevalence (86.3%), so this should be validated against the intended business population before deployment. The target is skewed toward churn rather than the more common low-churn setting. All results are conditional on this dataset.

1. Data audit and preparation

The source file contains 1,200 rows and 20 columns. Missing values were found in monthly_charges (30), total_charges (24), satisfaction_score (48), and monthly_usage_gb (36). No predictor rows were removed solely because a value was missing. Numeric values are median-imputed and categorical values are imputed with the most frequent category inside a scikit-learn pipeline.

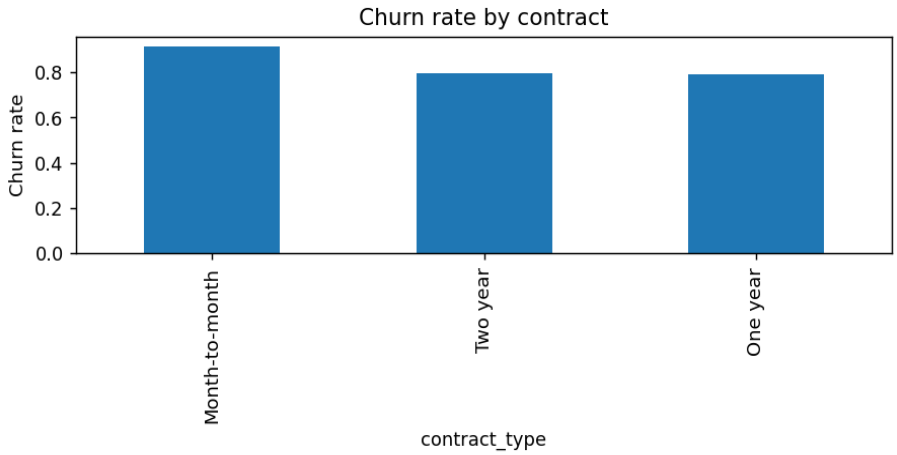
The target column is churn, encoded as 0/1. Its distribution is 164 non-churn and 1036 churn records. `customer_id` is retained for traceability but excluded from predictive inputs. Rows with invalid target labels would be excluded; the supplied file's target is valid for all retained records.

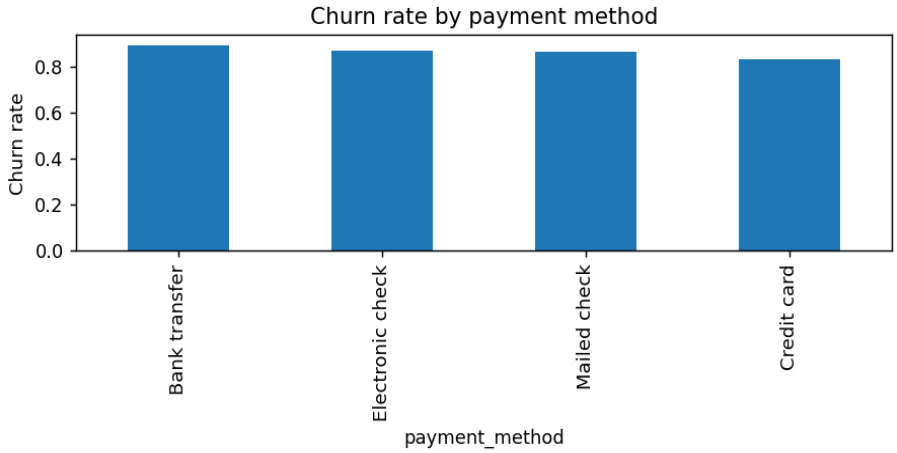


Cleaning decisions are explicit and reproducible. Extreme values are inspected through summaries and boxplots rather than deleted automatically because they may represent genuine customer behavior. The preprocessing pipeline fits imputers, scaling, and categorical encoding on training folds only.

2. Exploratory analysis

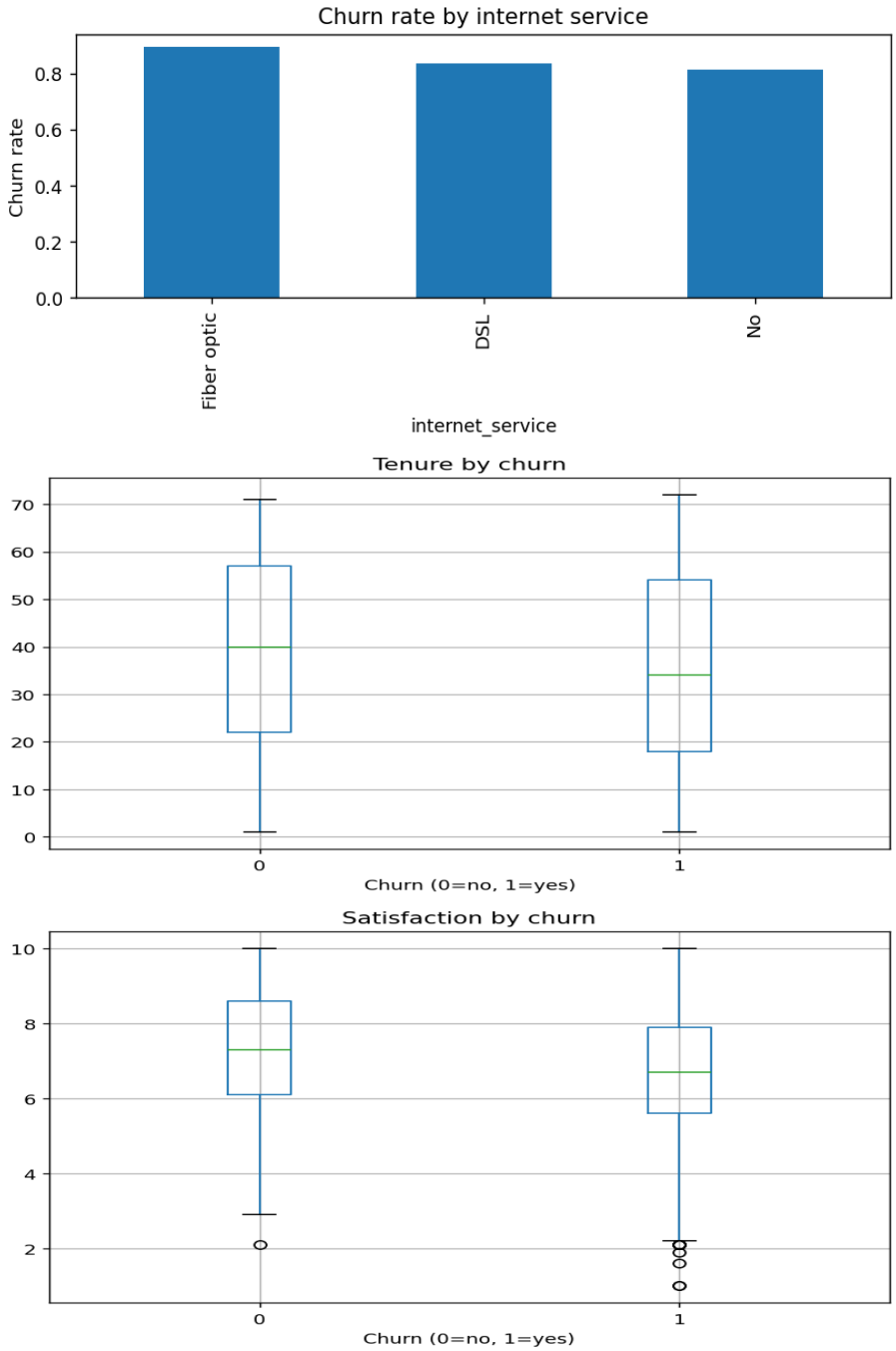
The charts below provide a first view of how churn differs across customer groups. These are descriptive comparisons and should be rechecked in later samples; they do not establish causation.





Contract type and payment method can inform segmentation. The business should investigate whether groups with higher observed churn have avoidable friction, pricing concerns, or service issues before designing a targeted intervention.

2. Exploratory analysis (continued)



The monthly-charge comparison is included in the notebook as an additional visualization. Review tenure, satisfaction, service type, and charges together: a single variable may be confounded by contract structure or customer segment.

3. Feature engineering and modelling

Eight features were added using input variables only: charges per tenure month; complaint intensity; payment-risk score (late payments plus electronic-check indicator); low-satisfaction flag; support-gap flag; usage per charge; discounted monthly charge; and short-tenure flag. These features represent business hypotheses and are not derived from the churn target, reducing the risk of target leakage.

The data was split into stratified 80% training and 20% test partitions with random seed 42. Numeric and categorical preprocessing was implemented using ColumnTransformer and Pipeline. Three classifiers were evaluated: Logistic Regression, Random Forest, and Extra Trees. GridSearchCV used three-fold StratifiedKFold and average precision as the model-selection score. Class weights were used instead of resampling.

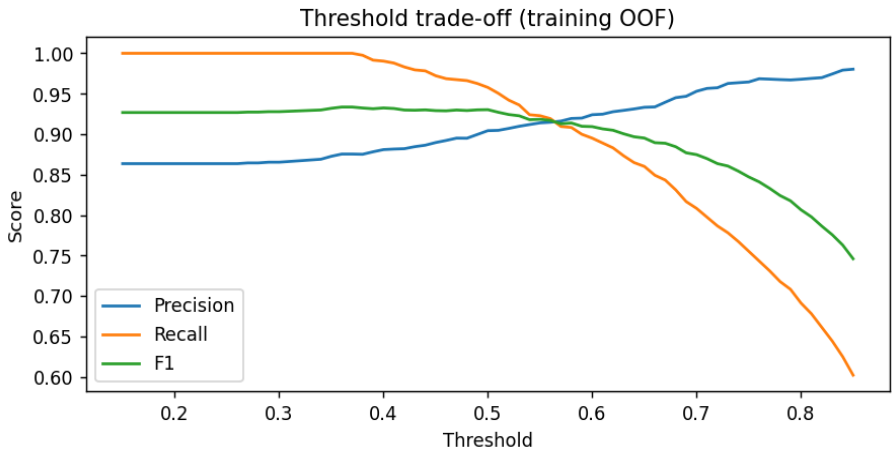
Model comparison on the untouched test partition

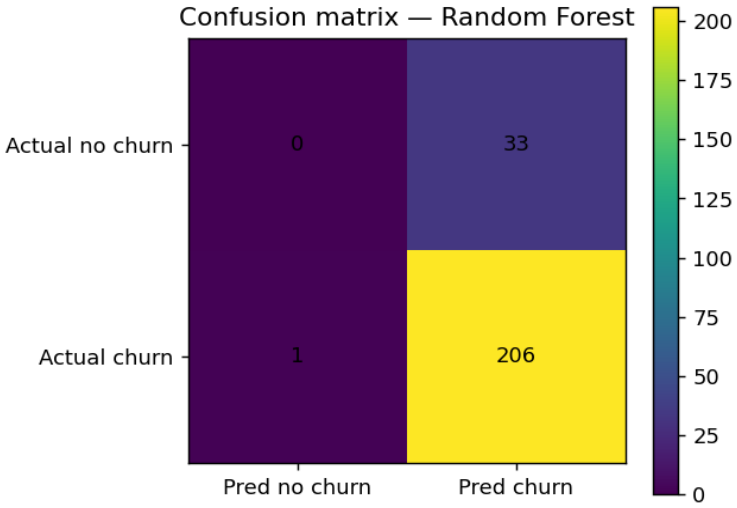
Model	CV PR-AUC	Precision	Recall	F1	ROC-AUC	PR-AUC	Accuracy
Random Forest	0.975	0.862	0.995	0.924	0.905	0.981	0.858
Extra Trees	0.949	0.896	0.961	0.928	0.872	0.976	0.871
Logistic Regression	0.966	0.973	0.855	0.910	0.900	0.975	0.854

The selected model is Random Forest because it achieved the strongest mean cross-validated PR-AUC. Its threshold (0.36) was selected from training out-of-fold probabilities by maximizing F2, which weights recall more heavily than precision. Test-set metrics were computed only after model and threshold selection.

4. Evaluation, calibration and explainability

At the selected operating threshold, test precision is 0.862, recall 0.995, F1 0.924, ROC-AUC 0.905, PR-AUC 0.981, accuracy 0.858, and Brier score 0.084. Precision estimates how many flagged customers are actual churners; recall estimates how many actual churners are found. Accuracy alone can hide poor minority-class detection and does not measure ranking quality.





Calibration should be reviewed before using predicted probabilities as literal likelihoods. The notebook calculates the Brier score and draws a calibration curve. The selected model's three CV PR-AUC fold scores should also be reviewed for variability. A model can rank customers well while its probabilities still need calibration.

Global and individual explanations

Permutation importance is used to estimate global predictive contribution by shuffling one raw input at a time and measuring the reduction in PR-AUC. This describes predictive association, not causation. Individual profiles below are high-risk test examples; the listed attributes are review cues, not proof of why a customer will leave.

Profile 1: predicted churn probability 99.9%; actual test label 1. Observed profile: tenure 34 months; satisfaction score 3.1; 1 complaints in the last 12 months; 0 late payments; Month-to-month contract; support plan=No, tech support=Yes. This is a review cue, not a causal explanation.

Profile 2: predicted churn probability 99.6%; actual test label 1. Observed profile: tenure 36 months; 0 complaints in the last 12 months; 1 late payments; Month-to-month contract; support plan=No, tech support=No. This is a review cue, not a causal explanation.

Profile 3: predicted churn probability 99.5%; actual test label 1. Observed profile: tenure 27 months; satisfaction score 5.5; 0 complaints in the last 12 months; 0 late payments; Month-to-month contract; support plan=No, tech support=Yes. This is a review cue, not a causal explanation.

5. Retention intervention simulation

The test set contains 240 customers. Contacting the top 20% means contacting 48 customers. The risk-ranked group contains 47 actual churners out of 207 test churners, capturing 22.7% of all churners in the test set. Randomly contacting 20% would capture an expected 41.4 churners on average. The model's lift is therefore modest in this test sample and should be interpreted carefully.

Illustrative cost-benefit assumptions

Contact cost = $5 per customer; value of a retained customer = $100; retention success rate among contacted churners = 25%. These are hypothetical assumptions, not values learned from the dataset.

Measure	Risk-ranked top 20%	Random 20% (expected)
Customers contacted	48	48
Churners in/expected in group	47	41.4
Expected retained customers	11.75	10.35
Contact cost	$240.00	$240.00
Expected retained-customer value	$1,175.00	$1,035.00
Net value under assumptions	$935.00	$795.00

The simulation assumes the intervention only has a chance of retaining customers who would otherwise churn, and does not account for margin differences, campaign fatigue, treatment effects, or operational constraints. The expected benefit is not a guarantee. The top-20% result should be compared with the random baseline and repeated on fresh data.

6. Recommendations, limitations and next steps

Evidence-linked retention actions

1. Prioritize the highest-risk segment for a human-reviewed outreach list, while monitoring precision, recall, and the number of customers contacted. The model is a prioritization aid, not an automatic decision-maker.

2. Review contract-related churn patterns and test clear renewal options or term-specific retention offers. Validate that the group differences persist before launching a campaign.

3. Review customer complaints and satisfaction signals; consider proactive service recovery for customers with unresolved issues. The dataset contains complaint and satisfaction indicators, but does not prove that an intervention will prevent churn.

4. Investigate payment friction among customers with late payments or particular payment methods. Offer reminders or simpler payment choices only where operationally and legally appropriate.

5. Test support-plan and technical-support offers for relevant customer segments using a controlled experiment to estimate incremental retention impact.

Limitations

The dataset is small (1,200 rows) and has unusually high churn prevalence. The test set is one random split, and the top-20% capture improvement over random is limited. Feature importance can be unstable when predictors are correlated. The financial simulation depends entirely on hypothetical assumptions. No model can prove causality from these observational records.

Recommended next steps

Validate schema and performance on newer, representative customer records; check probability calibration; evaluate performance across customer groups; run a controlled retention experiment; and revisit the operating threshold with real campaign capacity, costs, customer value, and success rates. Record model version, seed, package versions, and monitoring metrics before any production use.

Conclusion

The completed workflow meets the assignment's main technical requirements: data audit, six-plus visualizations, engineered features, leakage-aware pipelines, three tuned classifiers, threshold analysis, multiple evaluation metrics, permutation importance, customer profiles, calibration, robustness review, and a top-20% retention simulation. The best model in this run is Random Forest, but deployment should wait for validation on representative data.